APEX

Multi-Strategy Quantitative Trading Platform

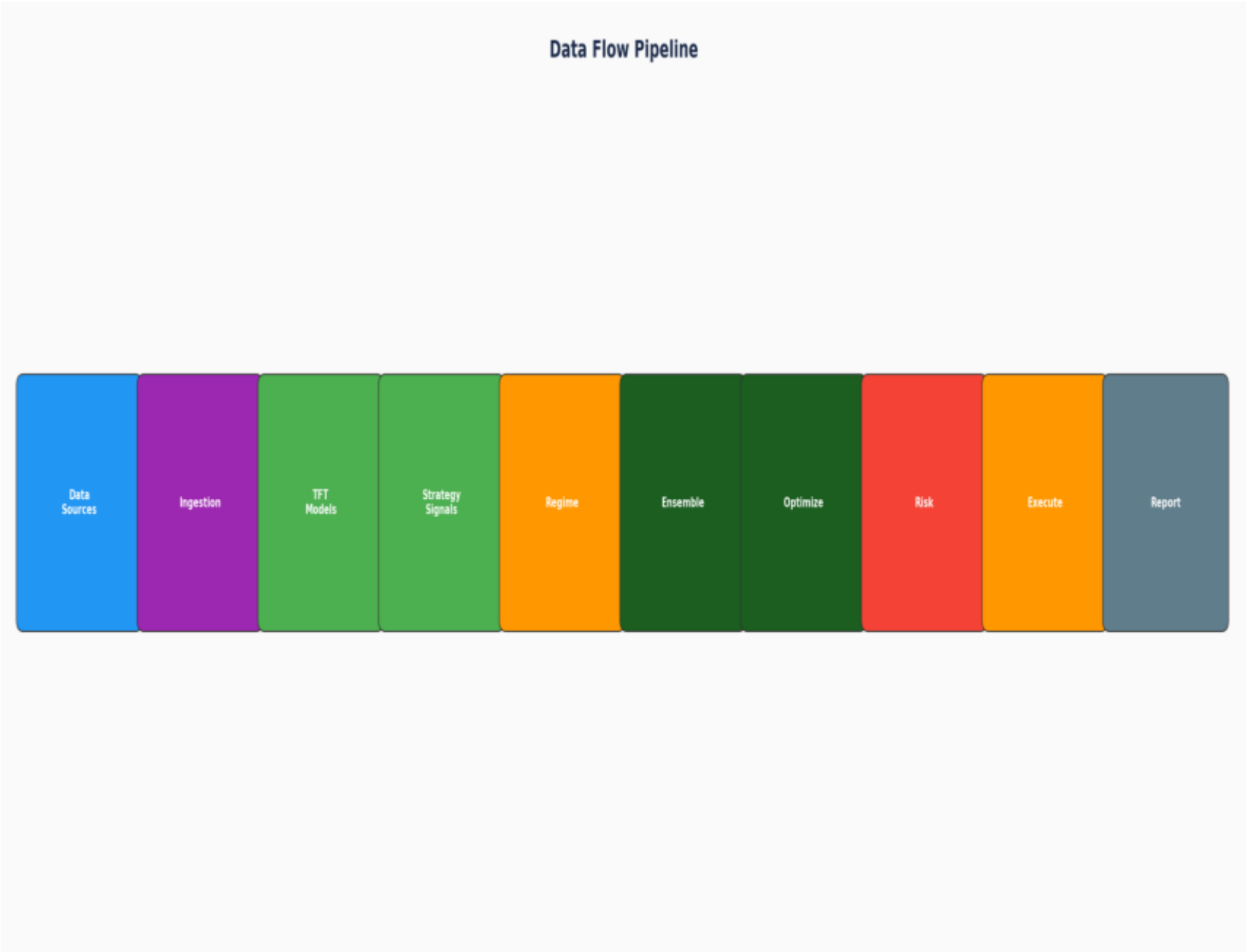
Complete Architecture Documentation

Sharpe 2.27	**11 Strategies**	**3 TFT Models**
107 Files	**31,943 Lines**	**6 Options Strategies**

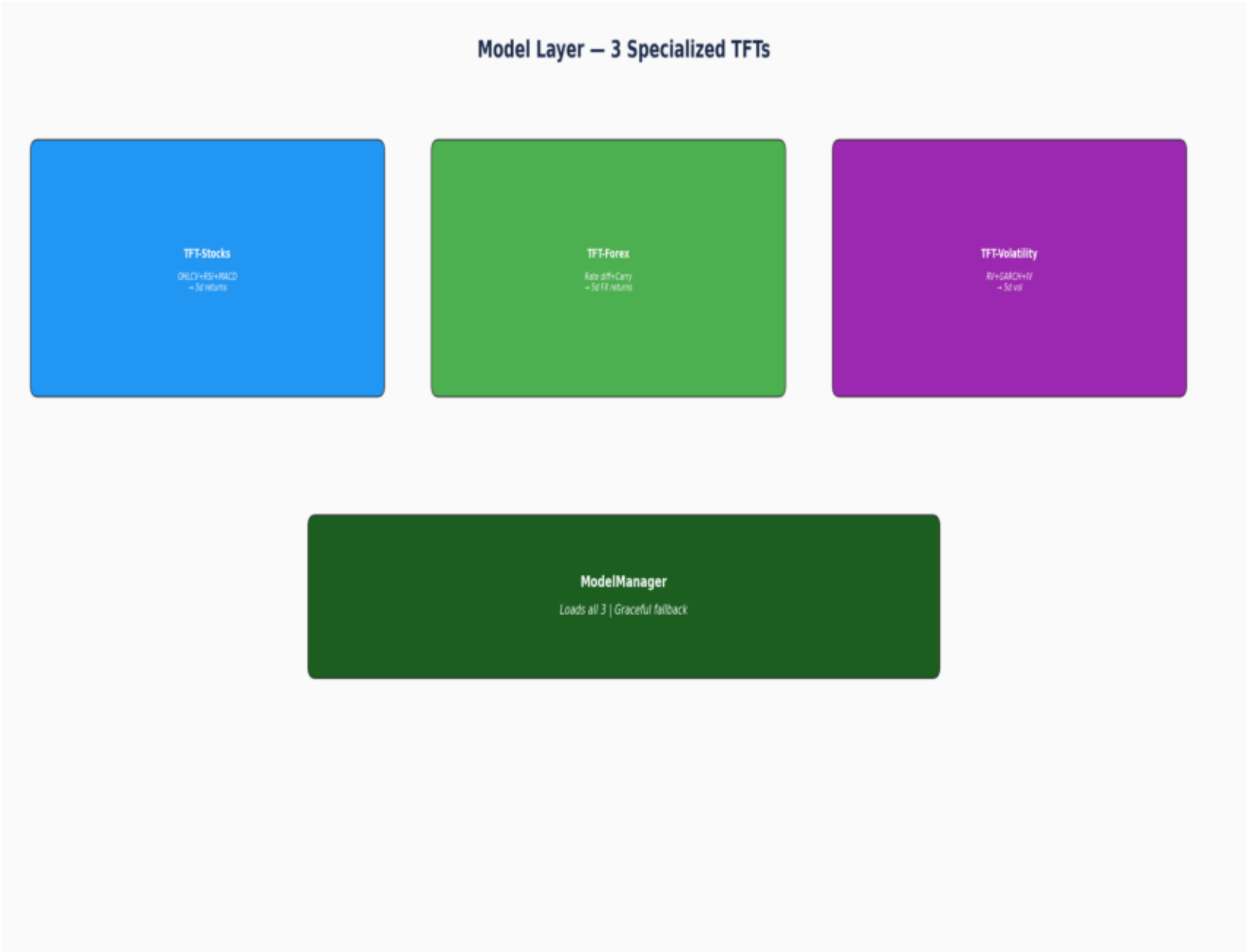
March 2026 | v3.0 Final | Confidential

System Architecture Overview

Data Flow Pipeline

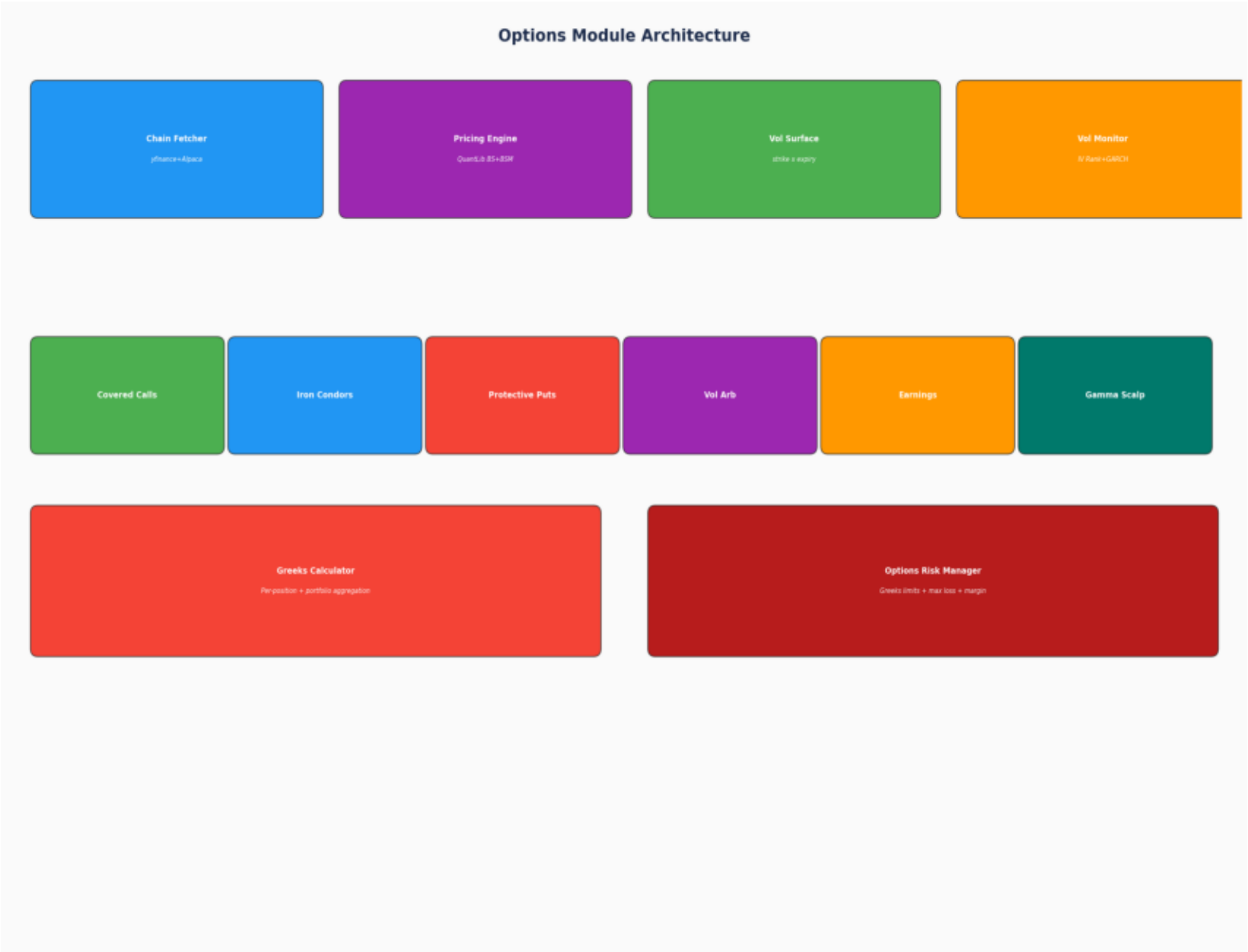


Model Layer — 3 Specialized TFTs



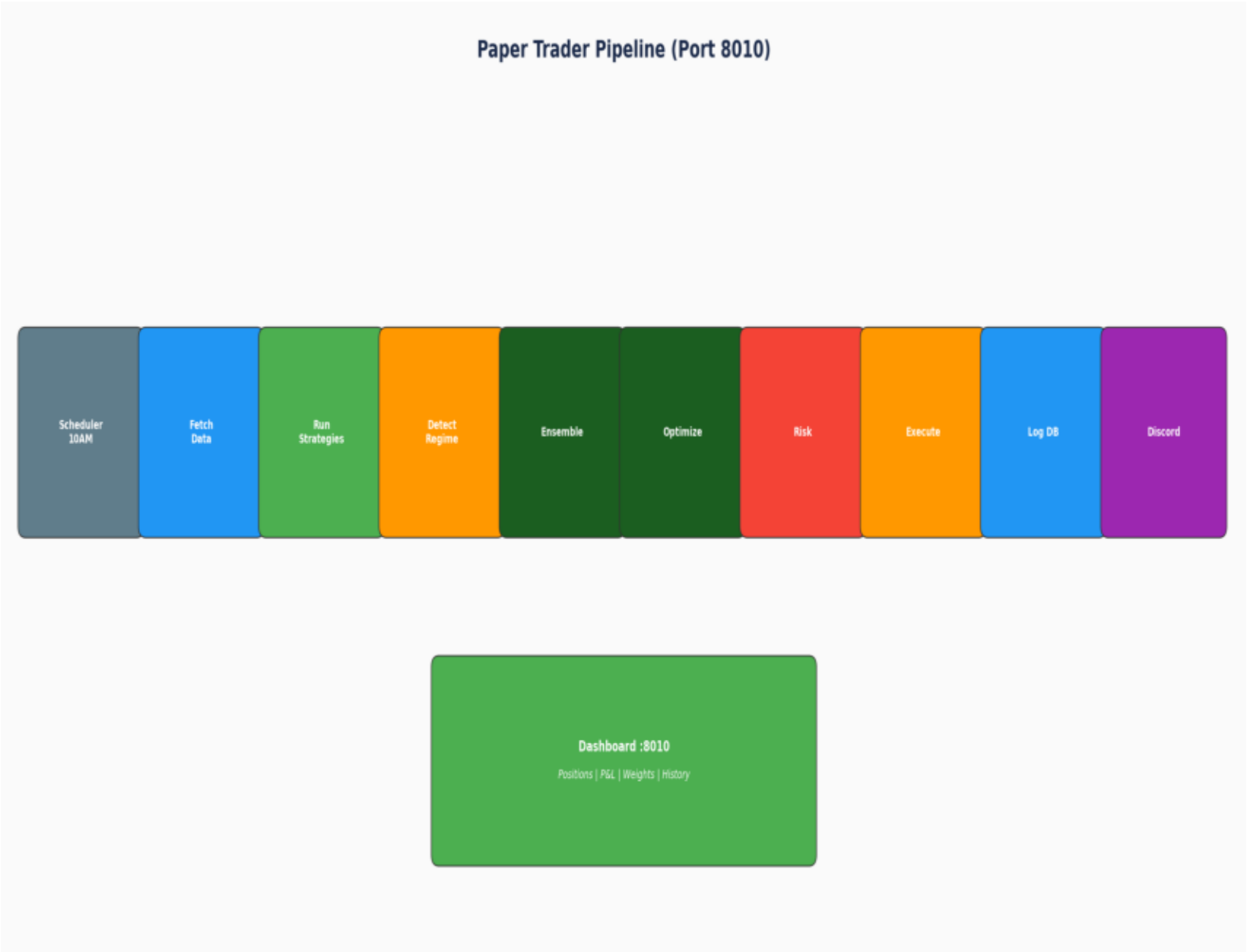
Strategy Engine — 11 Alpha Sources

Options Module Architecture



Risk Management — 5 Layers

Paper Trader Pipeline



Strategy Performance Summary

Strategy	Ann.Return	Sharpe	MaxDD	WinRate	Beta
Momentum (10 stocks)	+40.5%	2.270	6.2%	48.1%	0.28
Vol Arbitrage	+13.3%	2.047	7.2%	74.2%	N/A
Covered Calls	+22.2%	0.865	59.3%	66.3%	~1.0
Iron Condors	-2.9%	-0.821	7.6%	63.2%	~0
Momentum (29 stocks)	+1.1%	0.044	11.6%	48.1%	0.37
Pairs Trading	-1.9%	-1.419	5.1%	8.8%	0.001
FX Carry+Trend	-12.0%	-1.732	5.9%	42.3%	N/A

Portfolio Comparison

Portfolio	Ann.Return	Ann.Vol	Sharpe	MaxDD
Stocks Only (SPY)	+25.2%	17.9%	1.41	-18.7%
Stocks + FX	+21.5%	15.1%	1.42	-15.6%
Stocks + FX + Options	+15.9%	11.7%	1.36	-12.6%

Technology Stack

Deep Learning	PyTorch 2.10 + Forecasting 1.2 + Lightning 2.2
Factor Models	NumPy + Pandas + SciPy + Statsmodels
Options Pricing	QuantLib 1.41 + arch 8.0 (GARCH)
NLP/Sentiment	Transformers (FinBERT) + VADER
API Layer	FastAPI + Uvicorn + Pydantic
Message Bus	Apache Kafka
Database	PostgreSQL 15 / TimescaleDB
Cache	Redis 7
Broker	Alpaca API (aiohttp)
Model Registry	MLflow
Monitoring	Prometheus + Grafana
Containers	Docker + docker-compose (8+ services)
Language	Python 3.12 (107 files, 31,943 lines)